Demo: Meridian Works

Network summary report

Prepared on 28 October 2026

Question Where does work happen outside the formal structure?

Prepared for: a board. Groups and the whole network only; no member is shown, and groups of fewer than 5 members are pooled or left out.

40 members, 4 relationship layers analysed.

Ties are read in the direction they were rated (directed view).

Composite weights: set by the analyst.

Names are hidden: members appear as team, level and number codes, and names in free text are replaced.

This report describes the structure of working relationships. It is not an assessment of anyone, and should not be used to evaluate individual performance.

About the data

Coverage 95%, at or above the 80% threshold. Two of 40 people gave fewer than half of the ratings possible. What these two people's own ratings would show is missing; their results rest largely on other people's reports. Of the 708 ties reported where the other person also answered, 18% were not reported back.

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Executive summary

Prepared for a board: groups and the whole network only.

This summary covers 40 people in 5 groups by team or function and 4 kinds of working relationship. It describes patterns in how people say they work together, between groups rather than individuals. It is not an assessment of anyone. Each headline comes from a fixed rule applied to every group alike, and its questions are for discussion with the people involved.

Most of Finance's ties stay within Finance.

What we saw On informal collaboration, 54 of the 57 ties involving Finance's 8 members are between members of Finance, and 3 reach other groups (E-I index −0.89, where −1 means every tie stays inside). Six of its members receive fewer ties than most colleagues on most kinds of relationship. On connection strength, two of Finance's 8 members hold both of its ties with other groups (2 with Operations).

Why it may matter A group whose ties stay inside it can coordinate closely. Information, requests and ideas may also pass slowly between it and the rest of the organisation. When a group's links with others run through two people, those links depend on their time and availability.

Questions for discussion

- How does work reach Finance from the rest of the organisation, and how does it leave?
- Is this the pattern Finance and the groups it works with would expect, given the work they share?
- What would happen to Finance's links with other groups if one of those people were away?

Between Product and Sales, most formal collaboration has no informal counterpart.

What we saw Between Product and Sales, 85 of 119 formal collaboration ties (71%) come with no informal collaboration tie, counting each direction.

Why it may matter Formal ties carry the work as it is assigned. Without informal ties beside them, questions and exceptions may travel through formal channels, which can take longer.

Questions for discussion

- How do people in Product and Sales settle questions that fall outside formal processes?
- Does this match how people in Product and Sales describe working together?

Between People and Product, much collaboration happens outside the formal structure.

What we saw Between People and Product, 35 of 57 informal collaboration ties (61%)

come with no formal collaboration tie, counting each direction.

Why it may matter Informal ties often carry advice, coordination and favours that the formal structure does not show. Work that depends on them can be hard to see, and may change when people move.

Questions for discussion

– What work do the informal ties between People and Product carry?
– Is that work recognised in how People and Product are organised?

These patterns describe the network as the people in it reported it at one time. They are starting points for conversation with the groups involved, who will know what lies behind them.

Where teams connect

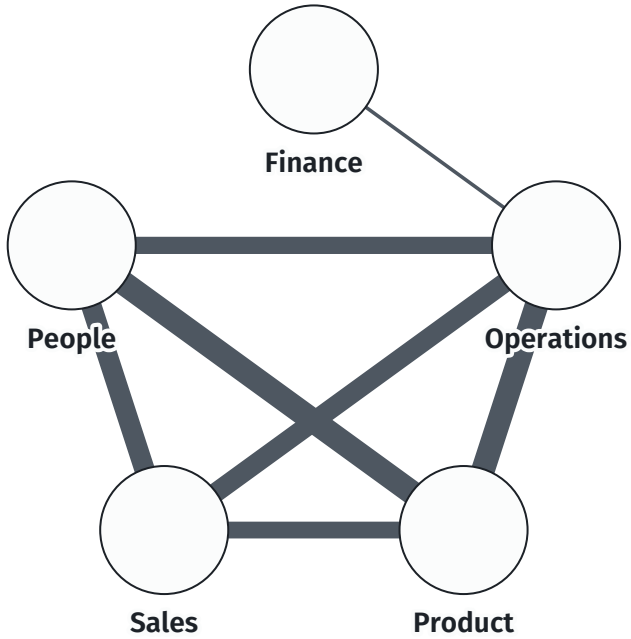
Question Where do teams connect, and where not?

Ask a colleague about Finance.

49% of ties stay within one team or function and 51% cross between them; of all possible pairs, 82% cross. E-I index +0.02 (−1: every tie within, +1: every tie across).

Product and People share the most ties between groups: 55. No tie joins Finance and Product, Finance and Sales, or Finance and People.

Groups, not members: ties drawn from Connection strength, within and between groups. The legend explains the picture.



Ties within and between groups, by team or function
Ties from Connection strength
Circle area: members in the group
Line width: share of possible ties between two groups, up to 43%

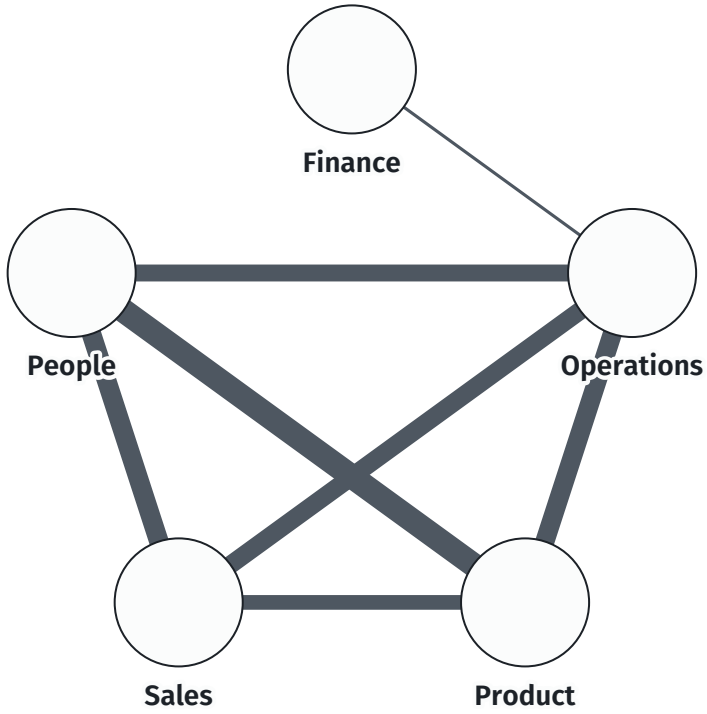
Work outside the structure

Question Where does work happen outside the formal structure?

Informal ties by team.

Of the 628 working ties, 16% are formal only, 17% informal only and 67% both. The largest share of informal-only ties is between People and Product: 35 of 58.

Groups, not members: ties drawn from Informal collaboration, within and between groups. The legend explains the picture.



Ties within and between groups, by team or function
Ties from Informal collaboration
Circle area: members in the group
Line width: share of possible ties between two groups, up to 46%

Data quality

Coverage is 95%, at or above the 80% threshold.

5,938 of 6,240 possible ratings were given.

Ratings	Value
Given	5,938
Declined	10
Not entered	292
Does not apply	0
Possible	6,240
Overall response rate	95%

The share of possible ratings that were given, across enabled layers. A rating of 0 counts as given; declined and never-entered ratings do not. A rating marked “does not apply” was never possible, so it is left out altogether.

2 raters are below 80%:

Where the ratings came from

Source	Value
Not recorded	5,948

Not recorded: the project does not say how these ratings were collected. Graticule records whether a rating came from a survey response, an imported file or entry in the matrix; ratings written by older versions or by other tools, and the demo’s ratings, carry no source. They are analysed like every other rating.

Survey response rates

No survey was run in Graticule for this project; ratings were imported or entered.

The method in brief

What the measures in this report mean, in a sentence or two each. They describe how people say they work together; none of them is a judgement of anyone.

How the data was gathered Each person rated their colleagues on a few kinds of working relationship, such as how often they collaborate or whom they ask for advice. A tie is a rating above the lowest point of a scale; a rating someone did not give is treated as unknown, never as no tie.

Kinds of relationship Each kind of relationship is analysed on its own, as a layer of the network. Where the report combines layers, it says which ones and how much each counts; the combination states the analyst's priorities and is not a finding.

Directions A tie has a direction: one person's rating of another may differ from the reverse. A report either counts each direction separately or reads the two directions of a pair together, and says which.

How dense the network is The share of possible ties that are present. A dense network is one in which most people are tied to most others.

Ties returned The share of ties that are given in both directions: the colleague named also names the person who named them.

Separate parts Groups of people with no route of ties between them. In most organisations everyone is connected, directly or through others.

Clusters Groups of people who are tied mostly to one another, found automatically from the ties alone. Modularity says how clear the clusters are; networks with clear clusters typically score between 0.3 and 0.7.

Ties within and between groups How many ties stay within a group, such as a team, and how many cross to other groups. The E-I index runs from −1, when every tie stays inside, to +1, when every tie crosses.

Concentration Whether ties fall evenly across people or gather on a few. Centralisation runs from 0, when everyone has as many ties, to 1, when every tie involves the same person.

Bridging How often a person lies on the shortest routes between colleagues. People with high bridging join up parts of the network that would otherwise be further apart.

Connections in and out How many colleagues name a person, and how many they name. These count ties; they say nothing about the quality of anyone's work.

Bridge ties and cut points A bridge tie joins two people who have no contact in common. A cut point is a person through whom alone some colleagues are linked to the rest of the network.

Demand on a few people The share of all requests, such as for advice, that the people named most often receive, set against the share they would receive if requests were spread evenly.

Formal and informal collaboration Formal collaboration is work that a role, process or reporting line requires; informal collaboration is work people choose to do together. Comparing the two shows where work runs outside the formal structure.

Negative ratings Some relationships are rated on a scale from negative to positive. Negative ratings are counted on their own and never offset by positive ones; a trio of people is balanced when none or two of their three ties are negative.

Insight rules Each pattern in the report was found by a fixed rule, with stated thresholds, applied to every group in the same way. A pattern is a question for the people involved, not a finding about anyone.

How complete the data is The share of possible ratings that were given. When coverage is below the threshold, figures about the whole network rest on incomplete data; people who gave few ratings are described largely by others' reports.

Appendix C gives the technical notes for the measures in this report; Graticule's method notes give every formula, with its references.

Appendix A. Key network metrics

Whole-network measures for the composite and each layer. Definitions and caveats are in the method notes.

Measure	Composite of all layers	Connection strength	Valence, positive ratings	Valence, negative ratings	Informal collaboration
Members	40	40	40	40	40
Ties	715	546	597	11	543
Density	0.458	0.350	0.383	0.007	0.348
Ties returned (tie reciprocity)	0.817	0.696	0.596	0.909	0.729
Average clustering	0.344	0.285	0.233	0.077	0.299
Separate parts (weakly connected components)	1	1	1	–	1
Members in the largest part (weak)	40	40	40	–	40
Centralisation of connections in	0.187	0.141	0.133	0.072	0.143
Centralisation of connections out	0.266	0.167	0.212	0.072	0.195
Centralisation of bridging	0.170	0.327	0.012	–	0.323
Communities (Louvain)	4	5	5	–	4
Modularity	0.457	0.460	0.378	–	0.414

Measure	Formal collaboration
Members	40
Ties	535
Density	0.343
Ties returned (tie reciprocity)	0.789
Average clustering	0.410
Separate parts (weakly connected components)	1
Members in the largest part (weak)	40
Centralisation of connections in	0.227
Centralisation of connections out	0.201
Centralisation of bridging	0.202
Communities (Louvain)	4
Modularity	0.497

For negative ties, "–" marks a measure that is not reported. Separate parts, communities, modularity and the centralisation of bridging describe how a network holds together, which negative ties do not: with few of them, nearly every member is a part and a community of their own.

Composite as calculated C = (0.33·r(Connection strength) + 0.33·r(Informal collaboration) + 0.33·r(Formal collaboration)) × (1 + 0.5·s(Valence)); C <= 1

Ties within and between groups by team or function, on Composite of all layers

Group	Members	Inside	Leaving	E-I index
Finance	8	56	6	−0.81
Operations	8	55	184	0.54
Product	8	56	256	0.64
Sales	8	56	244	0.63
People	8	56	182	0.53
All groups	40	279	436	0.22

E-I runs from −1 (every tie stays inside the group) to +1 (every tie leaves it). Larger groups have more chances of internal ties.

Structural balance: Valence

Measure	Value
Balanced trios	1,573
Unbalanced trios	79
Balance ratio	0.952

Formal and informal collaboration, pairs

Measure	Value
Pairs with formal ties only	101
Pairs with informal ties only	108
Pairs with formal and informal ties	419
Pairs with neither	819
Pairs not classified (a rating is missing)	113

Appendix B. Insight rules

Rule-based observations from this analysis. Each is a question to explore with the people involved, not a finding about anyone.

Rules about individual members (potential brokers, peripheral members, possible overload, negative clusters) are not part of outputs for this audience.

Only routes between groups

Rule Members are grouped by team or function. On the composite and on each unsigned layer except the conflict layers, with ties read without direction: a group of at least 3 members whose ties with every other group are held by at most 2 of its members, and by at most 50% of the group. The layer with the fewest such members is quoted. Also every member through whom alone at least 3 colleagues are linked to the rest of the network (a cut point).

On connection strength, two members of the 8 in Finance hold all 2 ties between Finance and other groups (2 with Operations). What would happen to Finance's links with the rest of the organisation if one of them were away?

Shown with Ties from connection strength, node size by bridging, grouped by team or function.

1 observation is not shown: it concerns a group of fewer than 5 members, or individual members.

Concentrated demand

Rule On the advice and workflow dependency layers, ties received are counted from the directed ratings in either view. Across the network: the 10% of members who receive the most ties (at least one member) hold at least 25% of at least 20 ties. Within each group by team or function of at least 5 members receiving at least 10 ties: the group's own top 10% hold at least 2.5 times the share they would hold if ties were spread evenly.

Not applied: this rule reads the advice and workflow dependency layers. Enable either layer and add its ratings.

Silos

Rule Members are grouped by team or function. A group of at least 3 members whose E-I index is −0.50 or lower on the composite or on any unsigned layer except the conflict layers, which means at least three ties inside the group for every tie outside it, counting only layers with at least 5 ties involving the group. The layer with the lowest index is quoted. Not applied when fewer than 2 groups have 3 members or more.

On informal collaboration, most ties involving Finance stay within Finance: 54 inside and 3 outside (E-I index −0.89). Six of its 8 members also receive fewer ties than most colleagues on most layers (see Peripheral members). How does work reach the rest of the organisation?

Shown with Ties from informal collaboration, node size by bridging, grouped by team or function.

Negative ratings not returned

Rule On each signed layer, negative ratings whose reverse rating is not negative: neutral, positive or not given. A layer with at least 3 is shown, with counts by team or function, the rater's group first. Read from the directed ratings in either view.

No observation: nothing in this analysis meets the rule.

Formal ties with no informal counterpart

Rule Members are grouped by team or function. For every pair of groups, and every group on its own: at least 50% of formal collaboration ties have no informal collaboration tie, and there are at least 5 such ties. Pairs with a missing rating on either layer are left out. The 5 members in the most such ties are highlighted.

Between Product and Sales, 85 of 119 formal collaboration ties (71%) have no informal counterpart. What happens between these people outside formal processes?

Shown with Ties from formal collaboration, node size by connections in, grouped by team or function, showing team or function Product or Sales.

Informal ties with no formal counterpart

Rule Members are grouped by team or function. For every pair of groups, and every group on its own: at least 50% of informal collaboration ties have no formal collaboration tie, and there are at least 5 such ties. Pairs with a missing rating on either layer are left out. The 5 members in the most such ties are highlighted.

Between People and Product, 35 of 57 informal collaboration ties (61%) have no formal counterpart. What work do these ties carry that the formal structure does not show?

Shown with Ties from informal collaboration, node size by connections in, grouped by team or function, showing team or function People or Product.

Caveats The thresholds are conventions chosen for this tool, not published standards; they are stated wherever a rule is shown so a reader can judge them. Every rule inherits the caveats of the metric it reads (sections 3 to 6): betweenness assumes shortest-path flow, the E-I index counts ties and favours larger groups, received strength depends on who answered. A negative rating not returned negatively may reflect a real difference in experience, a rater who rates everyone low or high, or a colleague who did not answer; the signed rule counts ratings, not people. The rules describe positions in the network, which have many causes (role, tenure, location, workload, the survey itself); an observation is a reason to ask, not an answer. Below the coverage threshold the panel repeats the coverage warning.

Appendix C. Technical notes

Drawn from Graticule's method notes, which give every formula in full with its references. Only the measures used in this report are included.

Not rated is not zero

A rating of 0 means "no connection" or "never" (on a signed layer, "neutral"). A missing rating means the person did not answer. The two are never treated alike: a missing rating does not create a tie, is not counted as a 0 in averages, and is counted as missing in data coverage. Inside the engine a missing rating is the value NaN; in project files it is null or an absent tie.

A third state, does not apply, records that the rater said the question has no answer for that colleague, such as the quality of a relationship they do not have. It is neither 0 nor missing. Metrics treat it like a missing rating (no tie), but coverage leaves it out of the ratings that were possible, because there was nothing to give. In project files it is null with not_applicable: true; in ties files and matrix entry it is written n/a.

Rescaling

Every layer is rescaled to 0–1 before any metric, so values from different layers can be compared and combined.

Unsigned layers: r = (v − min) / (max − min). On a 0–5 scale, 0 to 0, 1 to 0.2, 5 to 1.

Signed layers (valence, energy): s = v / max for v >= 0 and v / |min| for v < 0, so s runs from −1 to +1. Each signed layer is split into a positive sub-layer, w+ = max(s, 0), and a negative sub-layer, w− = max(−s, 0). The two are analysed separately and never netted.

Caveat: rescaling assumes the points of a rating scale are equally spaced. A "5" is treated as five times a "1".

Directed and symmetrised views

In the directed view A's rating of B and B's rating of A are separate ties.

In the symmetrised view each pair has one tie, combined from the two directions by the rule the analyst chooses:

mean: (w_ij + w_ji) / 2;

minimum: min(w_ij, w_ji), a tie only if both sides report one;

maximum: max(w_ij, w_ji), a tie if either side reports one;

both report a tie: min(w_ij, w_ji) when both are above 0, otherwise 0; missing when either direction is missing.

If only one of the two has rated the other, that rating is used on its own under mean, minimum and maximum (they then agree). Under "both report a tie" it is not: the pair is missing, so a member who gave no ratings has no ties at all in this view. If neither has rated the other, the pair is missing under every rule. Signed layers are combined before they are split, so when A rates B positively and B rates A negatively the pair's sign

follows the rule: the sign of the mean, negative under minimum, positive under maximum, and no tie under "both report a tie", which keeps a pair only when both ratings have the same sign (the one nearer zero).

The minimum rule is often described as "a tie only if both sides report one", and it is when both have answered; but because a missing direction is filled from the one given, a tie to someone who did not answer survives it. "Both report a tie" is the rule for confirmed ties only (build plan session 5; section 14).

Caveat: the minimum rule and "both report a tie" make the network sparser and the maximum rule denser; metrics can change markedly between them. "Both report a tie" turns non-response into absent ties, so below full coverage it removes real ties along with unconfirmed ones. The chosen rule is part of every result.

Composite

Meaning One combined relationship score per pair, weighting the layers by importance.

Caveats The composite is only as meaningful as the weights: it is a statement of priorities, not a measurement. Re-normalising over rated layers means a pair rated on one layer only is scored on that layer alone.

Density

Meaning The share of possible ties that exist.

Caveats Ignores strength. Not comparable across networks of very different size: large networks are sparser because people have limited time.

Reciprocity

Meaning How often ties are returned: when one member reports a tie to another, how often the other reports one back.

Caveats The two figures differ: a network with one mutual pair and one one-way pair has tie reciprocity 2/3 and dyad reciprocity 1/2. Both ignore tie strength. Missing ratings lower reciprocity, because an unanswered survey looks like an unreturned tie.

Average clustering

Meaning How clustered the network is overall: the mean of the members' local clustering.

Caveats Members with fewer than two contacts pull the average down.

Components

Meaning Groups of members connected by some chain of ties, with no ties to other groups.

Caveats An isolated member is a group of one.

Communities (Louvain) and modularity

Meaning Groups of members more densely tied to each other than to the rest, found automatically.

Caveats Louvain is a heuristic: it finds a good partition, not necessarily the best, and a different node order or seed can give a different one of similar quality. Graticule uses graphology's implementation, whose partitions can differ from NetworkX's; on every reference case its modularity is equal to or higher than NetworkX's. With a single run, or with graphology's default fast local moving, it fell up to 0.015 short. Modularity around 0.3 or above is usually read as clear community structure; values near 0 mean none. The resolution gamma = 1 is the same default as NetworkX; larger values find smaller communities.

Centralisation

Meaning How much the network is dominated by one member, compared with a star, the most centralised shape possible.

Caveats Computed for binary degree and betweenness only, where the star-graph maximum is established. Not defined for fewer than three members. NetworkX has no centralisation function; the reference values are computed from NetworkX degree and betweenness.

E-I index

Meaning Whether ties stay within groups (such as teams) or cross between them.

Caveats Counts ties, not strength. Larger groups have more internal opportunities, so their index is lower even when members mix freely; compare with cross-group density. Members with no value for the attribute are left out, and the number left out is reported.

Structural balance

Meaning Whether trios of members are in a stable pattern. Balance theory (Heider, 1946; Cartwright & Harary, 1956) holds that "my friend's friend is my friend" (+++) and "my enemy's enemy is my friend" (+−−) are stable, while a trio with one negative pair (++−) or three (−−−) is under strain.

Caveats Trios with a neutral (0) or unrated pair are not counted. The dyad sign depends on the symmetrisation rule when the two directions disagree.

Formal–informal comparison

Meaning Classifies every pair as formal only, informal only, both, or neither, using the formal and informal collaboration layers.

Caveats "Formal only" means the collaboration is required but not reported as happening outside formal channels; it is a prompt for inquiry, not evidence of a problem.

Data coverage

Meaning The share of possible ratings that were given, per rater and overall.

Ethics statement

This report maps working relationships between named people. It describes the structure of a network; it does not measure anyone's performance, and it should not be used in performance evaluation or in decisions about individuals.

Participants should have given informed consent, and should have been told what would be collected, who would see the results and how the results would be used. Survey responses are confidential, not anonymous: the analyst sees who gave which ratings.

Processing named employee data is likely to engage data protection law. Store, share and delete this report as personal data, following the advice your organisation has taken.

Every position in a network has many causes, including role, tenure, location, workload and the survey itself. Observations here are prompts for conversation with the people involved, not conclusions about them.

Names in this report are replaced by team, level and number codes. Codes of small teams can still identify someone, so treat the report as personal data.